

Audit of a Five-Dimensional KDE for the $n = 3$ Finite-Time Total-Entropy Fluctuation Theorem

Numerical validation report

2 September 2026

Verdict

The frozen five-dimensional KDE does *not* resolve a reliable finite-time total-entropy fluctuation-theorem (FT) test for the driven $n = 3$ chain at $(T_L, T_R) = (10, 2)$ and block time $t = 20$. This is a negative result about the density-estimation route, not evidence that the physical FT is false.

The conclusion follows before inspecting any total-entropy slope: the KDE fails the exact equilibrium Gibbs-density accuracy gate at both $T = 6$ and $T = 10$, two independent driven KDEs disagree beyond every frozen tolerance, and 42 driven endpoint pairs have zero density under the frozen binned kernel. Under the predeclared no-extrapolation rule, total-entropy sign counts, the DFT slope, and the integral FT (IFT) were therefore not computed.

1 Data and immutable protocol

All samples use the projection-free Cartesian sampler with $n = 3$, $\gamma = 0.1$, $\Delta t = 5 \times 10^{-4}$, burn-in 500, block length 20, 128 independent streams, and 7813 nonoverlapping blocks per stream. Each case contains 1,000,064 blocks. The driven seed is 2026083133; equilibrium seeds are 2026090106 ($T = 6$) and 2026090110 ($T = 10$).

The offloaded driven CSV was deterministically restored with the original source and seed. Its decompressed SHA-256 is

4f728b3d0e007d704d90734b0888c00ec05b60f09385c6cfd079f3417d7a088f,

exactly matching the accepted original sample. Hence this audit did not replace the driven data with a statistically different run.

2 KDE construction

The reduced state is

$$x = (I_1, I_2, I_3, \theta_1, \theta_3), \quad z = (\log I_1, \log I_2, \log I_3, \theta_1, \theta_3).$$

The estimator is a product Gaussian KDE binned on a $48^3 \times 32^2$ grid. The action axes use ordinary Gaussian convolution; the angles use wrapped Gaussian convolution on $[-\pi, \pi)$. The density with respect

to the physical reduced measure includes the transformation Jacobian,

$$\log \rho_x = \log \rho_z - \sum_{j=1}^3 \log I_j.$$

Whole streams, not individual blocks, define two cross-fitting folds. Each endpoint is evaluated by a KDE trained only on the opposite stream fold. Scott’s five-dimensional factor uses the correlation-adjusted effective size $N_{\text{eff}} = N/g$, where g is the largest initial-positive-sequence statistical inefficiency among the three log actions and the sine/cosine of both angles. Here g lies between 1.000 and 1.016 for the primary folds, so $N_{\text{eff}} \simeq (4.92\text{--}4.98) \times 10^5$. Representative action bandwidths are 0.294–0.316 for the driven sample and 0.296–0.300 at equilibrium; angular bandwidths are 0.398–0.423 radians. No bandwidth multiplier or FT fit window was tuned.

For each supported block,

$$\Delta s_{\text{sys}} = -\log \rho_x(x_{\text{end}}) + \log \rho_x(x_{\text{start}}), \quad \Delta s_{\text{tot}} = \Sigma^m + \Delta s_{\text{sys}}.$$

3 KDE accuracy: exact equilibrium test

This section is reported before any driven FT number. At equilibrium, $\log \rho_{\text{exact}} = -E/T + \text{constant}$ and therefore

$$\Delta s_{\text{sys}}^{\text{exact}} = \frac{E_{\text{end}} - E_{\text{start}}}{T}.$$

Table 1 compares held-out KDE errors with thresholds frozen before the analysis. Unsupported points are omitted from the numerical RMSE but reported explicitly; their presence itself forces failure.

Table 1: Equilibrium known-answer KDE gate. The four frozen maxima were 0.15, 0.10, 0.25, and 0.50, respectively.

case	unsupported /1,000,064	endpoint RMSE	increment RMSE	tail 1% RMSE	increment $q_{.99} e $	gate
$T = 6$	52	0.273456	0.387016	1.70194	1.38458	FAIL
$T = 10$	62	0.270944	0.383225	1.72094	1.38283	FAIL

Every primary error is well above its tolerance. The increment RMSE grows from 0.2982 ($T = 6$) and 0.2925 ($T = 10$) in the lowest-energy 80% to 1.7019 and 1.7209 in the lowest-density one percent. The tail error is therefore not a small uniform normalization offset; it directly affects the endpoint log-density ratios required by total entropy.

4 Driven KDE stability and support

Two additional KDEs were trained on disjoint stream groups and evaluated on the same 328,146 driven audit blocks. Twenty-six blocks lack common support. On the 328,120 supported blocks, the centered endpoint log-density RMSE between the two estimators is 0.220699 (limit 0.15), the system-entropy increment disagreement RMSE is 0.311976 (limit 0.10), and the latter rises to 1.46711 in the lowest-density one percent (limit 0.25). Its 99th absolute percentile is 1.00570 (limit 0.50). Thus the independent-estimator gate also fails.

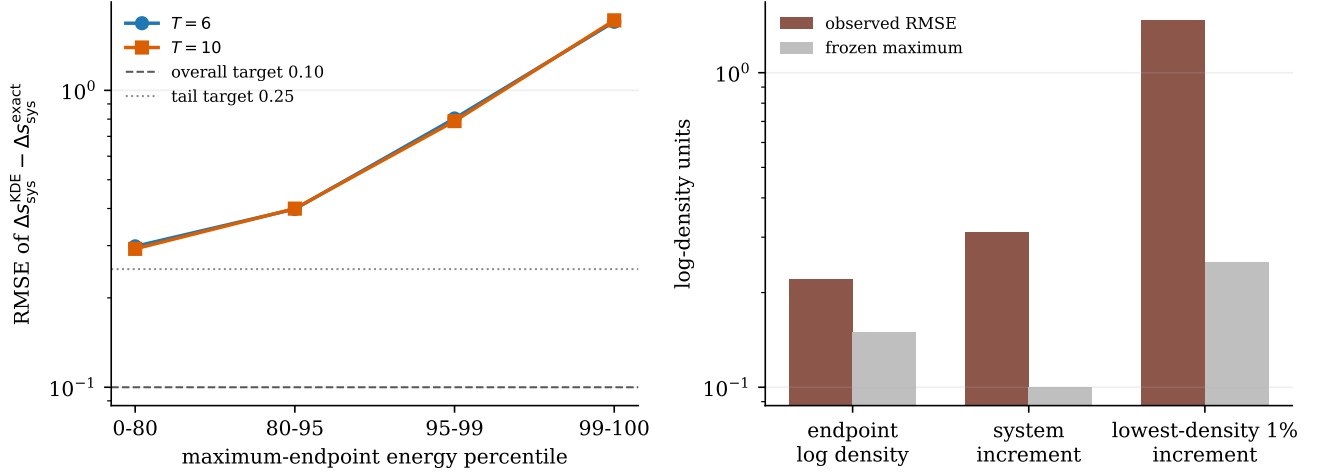


Figure 1: Left: equilibrium system-entropy increment error grows strongly into the low-density tail. Right: disagreement between two independently trained KDEs on the same held-out driven blocks also exceeds the frozen limits.

In the primary two-fold fit, 1,000,022 of 1,000,064 endpoint pairs are supported and 42 are unsupported. On the supported subset only, the estimated Δs_{sys} has mean -9.09×10^{-6} , standard deviation 1.66178, range $[-24.1348, 26.8440]$, and one/99-percentiles $[-4.07981, 4.08584]$. These values are descriptive: removing the 42 unsupported tail blocks would bias exactly the rare-event statistics needed for the FT.

5 What can be reported from the driven sample

Medium entropy, which does not use the KDE

There are 41 negative medium-entropy blocks among 1,000,064:

$$\hat{P}(\Sigma^m < 0) = 4.0997376 \times 10^{-5}, \quad 95\% \text{ stream-bootstrap CI} = [2.8998144, 5.3996544] \times 10^{-5}.$$

The frozen Freedman–Diaconis bin width is 0.47850865. No symmetric bin pair has 20 observations on both sides, so no medium-entropy symmetry slope is available. Table 2 lists every bin containing at least one negative observation; these raw counts sum to 41.

Table 2: Raw medium-entropy symmetric-bin counts with $N(-a) > 0$.

bin	a interval	center	$N(+a)$	$N(-a)$
0	$[0, 0.4785)$	0.2393	9	13
1	$[0.4785, 0.9570)$	0.7178	13	4
2	$[0.9570, 1.4355)$	1.1963	12	3
3	$[1.4355, 1.9140)$	1.6748	8	6
4	$[1.9140, 2.3925)$	2.1533	21	7
5	$[2.3925, 2.8711)$	2.6318	38	1
6	$[2.8711, 3.3496)$	3.1103	39	2
8	$[3.8281, 4.3066)$	4.0673	69	2

bin	a interval	center	$N(+a)$	$N(-a)$
10	[4.7851, 5.2636)	5.0243	81	1
13	[6.2206, 6.6991)	6.4599	121	1
15	[7.1776, 7.6561)	7.4169	151	1

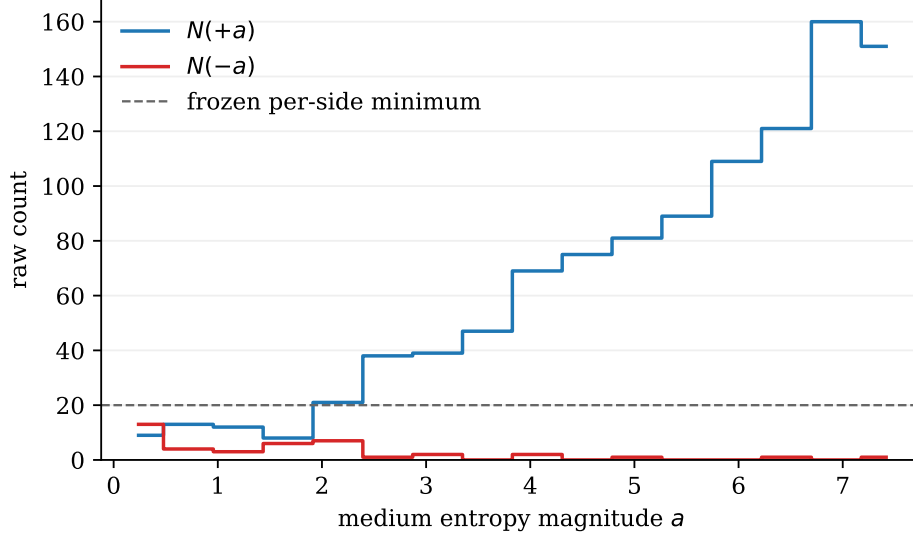


Figure 2: Raw two-sided medium-entropy support. Neither side reaches the predeclared minimum of 20 in the same bin.

The medium-only exponential diagnostic is also unresolved:

$$\log\langle e^{-\Sigma^m} \rangle = -5.7735605, \quad 95\% \text{ CI} = [-8.0873157, -4.8624448].$$

Its exponential-weight effective sample size is only 2.2880 out of 1,000,064; one block contributes 61.55% of the sum and deleting one stream can change the log average by 0.9481. This number is not a resolved IFT test, and medium entropy is not the exact finite-time total entropy.

Total entropy

The total-entropy entries are intentionally recorded as “not computed”:

quantity	result
$\#\{\Delta_{s_{\text{tot}}} < 0\}$	not computed (42 unsupported pairs)
$P(\Delta_{s_{\text{tot}}} < 0)$	not computed
DFT slope and confidence interval	not computed
$\log\langle e^{-\Delta_{s_{\text{tot}}}} \rangle$	not computed

Filling these points with a density floor, an infinite-kernel tail, or plus-four pseudo-counts would manufacture precisely the negative-tail evidence the audit is designed to test.

6 Integrity and provenance

All three input archives pass Zstandard integrity checks and their decompressed hashes match the pre-declared values. Every archive contains 1,000,064 ordered finite rows; consecutive saved endpoints agree exactly at output precision. The driven identity $\Sigma^m = -Q_L/10 - Q_R/2$ has maximum absolute recomputation error 2.84217×10^{-14} . The restored simulator reports zero midpoint failures and first-law residual RMS rate 1.48691×10^{-5} .

item	SHA-256 / identifier
frozen scientific protocol	commit 0b53e573afdd324fa44b4c7cb50c8bc579d40838
unsupported-point handling	commit 0faec919ad58ca8504a6d08a660f1c3492feadfc
CSV implementation fix	commit 4e423a965ae076039f7aa3b37fac8b3842b93b1f
simulator source	9ae5835ed708c8794c8b00ba799b23761482953aaf0ed47cd0b4ba3966d4eaf2
simulator binary	93c4aa6d046f3c35cd0ea1136091fc44ef1b64baf6237e4663ca9e86b995a156
analysis source	59ddfdb9d530116be169a9ce2651b44580f9c7cc666c7a4f4435e9728dc80e7c
driven compressed data	5979d35dd335574bc01add81d4c8d97759151a62f4fd6acadc111eb6bdccd7a2

The exact restoration command was

```
entropy_ft_n3_eq sample_n3 10 2 3 8 500 20 7813 0.0005 \
  2026083133 8 raw/driven 1
```

and the final analysis command was

```
python3 analyze_kde_ft.py --driven raw/driven_blocks.csv.zst \
  --equilibrium-t6 ../entropy_ft_n3_equilibrium_2026-09-01/raw/T6_blocks.csv.zst \
  --equilibrium-t10 ../entropy_ft_n3_equilibrium_2026-09-01/raw/T10_blocks.csv.zst \
  --output analysis
```

Claim boundary

This experiment establishes that the specified cross-fitted five-dimensional KDE is insufficiently accurate in the endpoint tails to extend the exact finite-time total-entropy FT from $n = 2$ to $n = 3$. It neither validates nor falsifies the physical $n = 3$ FT. A future total-entropy claim requires a stationary-density or density-ratio estimator that passes the exact Gibbs tail audit and yields nonzero, stable support for every driven endpoint used in the two-tail statistic.